

Rail fence cipher: We basically write the plaintext in a way which creates linear patterns in a spiral way. For example: if you have a look at your (US english) keyboard, you would see that the character sequence “QAZSEDCFTGBHUJMKOL” are in a spiral sequence. But when you read them in order the letters appear in rows from top to bottom, it becomes “QETUOASDFGHJKLZCBM”. Now we use the same technique to encrypt “ILOVEDIGITMAGAZINE”. It would look like:

I E I G N
 L V D G T A A I E
 O I M Z

Reading it on rows, it becomes “IEIGNLVDGTAAIEOIMZ”. In this case, the key is ‘3’ as we are using 3 rows to do the transformation. Try to do it with key of ‘4’ and you would find that the ciphertext changes.

Route Cipher: In this case, the plaintext is laid out in a grid and then written as a sequence of characters which is derived from a visible sequence over the grid. Let’s assume the same plaintext and lay it in vertical columns on a 6 x 3 grid:

I	V	I	T	G	I
L	E	G	M	A	N
O	D	I	A	Z	E

Now read it “inward spiral beginning top right corner”. The ciphertext becomes “IGTIVILODIAZENAMGE”. However, if you’re using a grid of 5 x 4, the grid would become like:

I	E	I	G	N
L	D	T	A	E
O	I	M	Z	
V	G	A	I	

The last two cells on the lower rows remain empty. Those can be filled with any nonsense which the interpreter on the other end can easily understand and discard. In this case the width of the grid can serve as the key for the encryption. The number of rows can be calculated using simple mathematics given the padding was done.

Columnar Transposition: This technique is a bit similar to the router cipher. Here too we lay out the plaintext in columns but the way we would